

Troubleshooting and Challenges Encountered

Several parts of this project required troubleshooting beyond basic configuration. Most issues were not obvious at first and required digging into logs, IAM permissions, and AWS service interactions to understand what was actually happening.

CloudWatch Agent Credential Errors

- Disk and memory metrics initially failed to appear in CloudWatch.
- Agent logs showed credential-related errors.
- Cause: EC2 instances had no IAM role attached.
- Resolution: Created and attached an IAM role with CloudWatchAgentServerPolicy, then restarted the agent.

Lambda Logs Not Appearing in CloudWatch

- No log streams appeared despite GuardDuty findings.
- Cause: Lambda had never been invoked.
- Resolution: Verified execution role, manually invoked Lambda, confirmed log stream creation.

EventBridge Rule Matching Issues

- EventBridge rule showed no matched events.
- Cause: Event pattern too strict.
- Resolution: Relaxed pattern, verified invocation, then refined rule.

Region Mismatch Across Services

- Events failed to trigger Lambda despite correct configuration.
- Cause: Services were operating in different regions.
- Resolution: Aligned EC2, GuardDuty, EventBridge, Lambda, and CloudWatch to the same region.

GuardDuty Detection Timing Confusion

- Expected immediate findings after port scan.
- Cause: GuardDuty analysis delay.

- Resolution: Allowed 30–60 minutes for findings to appear.

Parsing Alarm Data in Lambda

- Alarm payloads used different identifiers.
- Cause: Native EC2 vs CloudWatch Agent metric differences.
- Resolution: Added logic to handle InstanceId and host dimensions and map hostnames to instance IDs.

Key Takeaway from Troubleshooting

Most issues were caused by service interactions—permissions, regions, event patterns, and timing—rather than syntax errors. Resolving them required careful log analysis and validating each component individually.